Crypto Lab 1 - Shishir Vuppala

Challenge 1: Two-Time Pad

flag: `cs409{you_should_always_think_twice_before_reusing_one_time_pad_key!}`

Solution

Because we have used the same key twice, We have the following relation

$$c_! \oplus c_2 = (m_1 \oplus k) \oplus (m_2 \oplus k) = m_1 \oplus m_2$$

So assuming m_1 is the flag, if we can figure out m_2 then we can xor it with $c_1 \oplus c_2$ to get flag. The information we have is that m_2 is a valid english sentence and so there is some structure because that english sentence should make sense. (So a sentence like "I are ..." is not allowed and so after 'I' the most likely candidates are "am" or "will"). So we guess the next words based on previous words decoded so far. This technique is called crib dragging apparently. The TAs said we could use a website for this, however, I didn't read that in time, and so, coded up a python file to allow me to efficiently crib drag. It allowed me to add text to wither flag or m_2 and it would give me the corresponsing text on the other and if the text on both made sense then I could confirm my choice and move on.

Since we know that the flag starts with 'cs409' we can use that to start off our crib drag. This showed that the starting of m_2 is 'Crypto' which then obviously meant next few letters of m_2 are 'graphy ' and like this we continue. I got stuck for a bit at 'Cryptography has gone from a h' so then googled it thinking it was a quote and I found a line in a cryptography book which had a similiar line (not the same though) and had 'heuristic' in it and so using that i continued my crib dragging.

The session log of my crib dragging is present in this folder as `session_log.txt`

Code

The script used to solve this challenge is written in `decrypt.py`